

[2] Project Architecture

The diagram illustrates the project architecture, showing the interconnection of various hardware modules. The modules are represented by colored boxes, and their connections are shown by lines with labels.

Power (Red box) connects to **MCU - Power** (Green box) via **File: Power.kicad_sch**.

Sensing - VBAT (Pink box) connects to **MCU - IO Domain** (Green box) via **File: sense_temp.kicad_sch**.

Sensing - Temperature (Pink box) connects to **MCU - IO Domain** (Green box) via **File: sense_vbat.kicad_sch**.

Sensing - Position (Pink box) connects to **MCU - IO Domain** (Green box) via **File: sense_pos.kicad_sch**.

Interface - Interconnects (Cyan box) connects to **MCU - IO Domain** (Green box) via **File: interconnects.kicad_sch**.

Interface - FDCAN Tran. (Cyan box) connects to **MCU - IO Domain** (Green box) via **File: fdcan.kicad_sch**.

MCU - Power (Green box) connects to **MCU - IO Domain** (Green box) via **File: mcu-power.kicad_sch**.

MCU - IO Domain (Green box) connects to **FOC - Gate Driver** (Yellow box) via **File: gate-driver.kicad_sch**.

MCU - IO Domain (Green box) connects to **FOC - Inverter** (Orange box) via **File: inverter.kicad_sch**.

Interface - Status LEDs (Cyan box) connects to **MCU - IO Domain** (Green box) via **File: status-led.kicad_sch**.

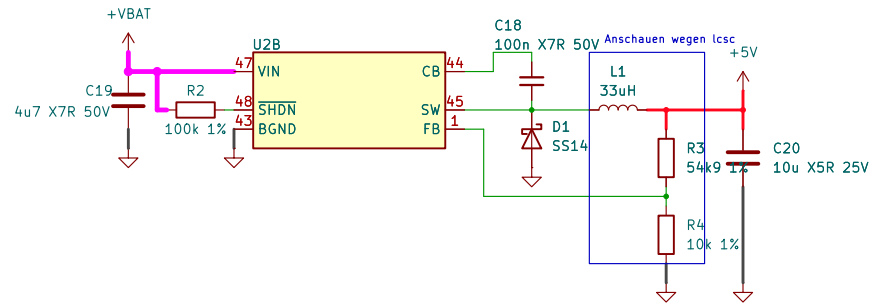
MISC. (Blue box) connects to **MCU - IO Domain** (Green box) via **File: misc.kicad_sch**.

Comments:	Designer:	Date:	Company:
	Moritz Heinze	2026-09-07	ATLAS ROBOTIC SYSTEMS
Board Name: ATLAS Motion			
Sheet Title:	File Name:	Git Hash:	Revision:
Project Architecture	atlas-electronics.kicad_sch	as89s1	0.0.1
Sheet Path:			Size:
/			A4
			Sheet:
			2 of 13

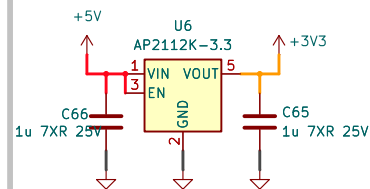
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2 of 13

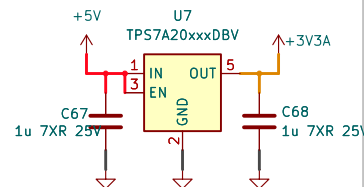
[3] Power



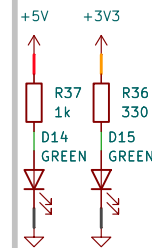
LMR16006X in Gate-Driver and LDO



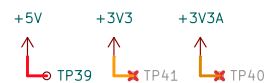
+3V3 LDO



+3V3A LDO



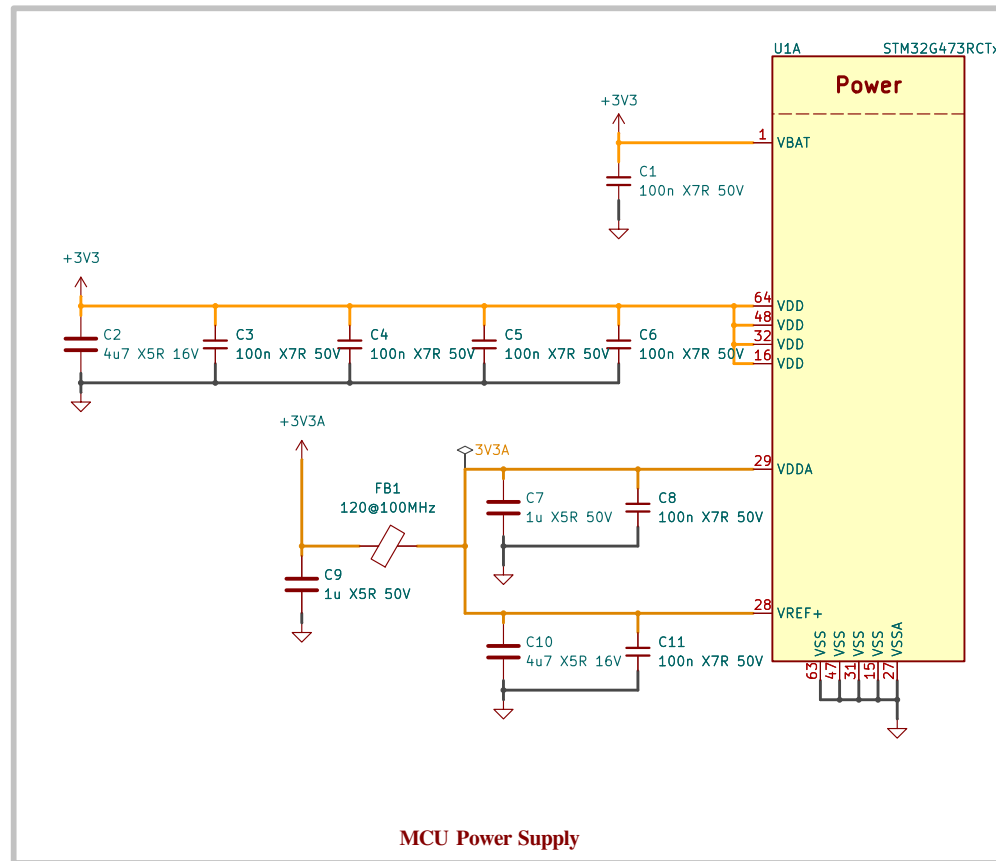
Power LED



Testpoints

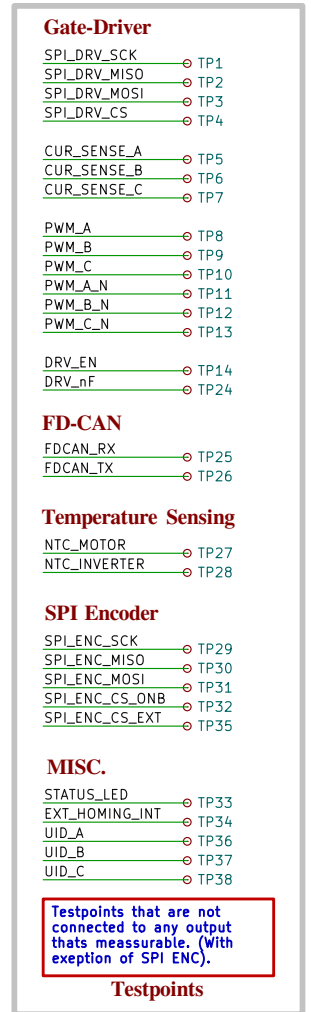
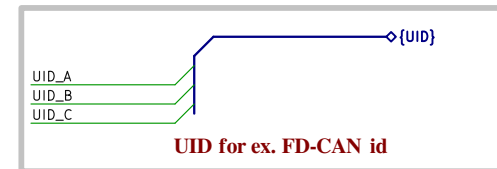
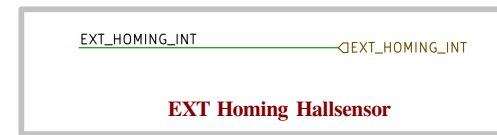
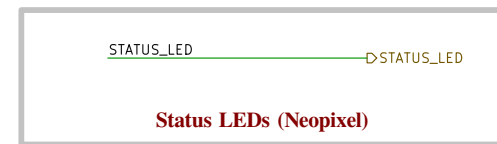
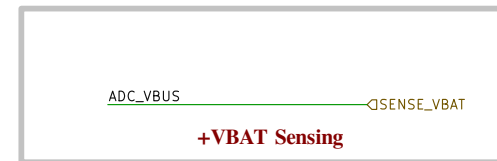
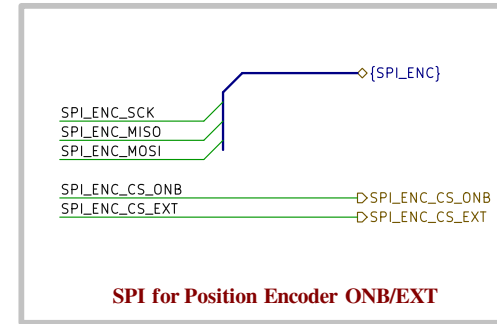
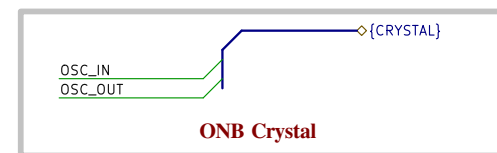
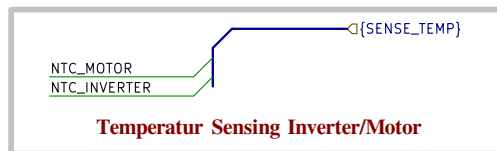
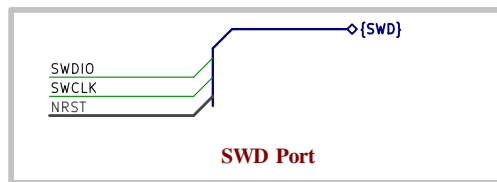
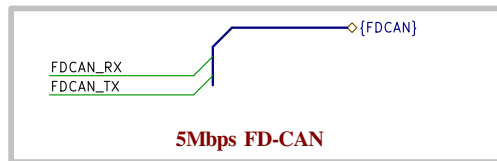
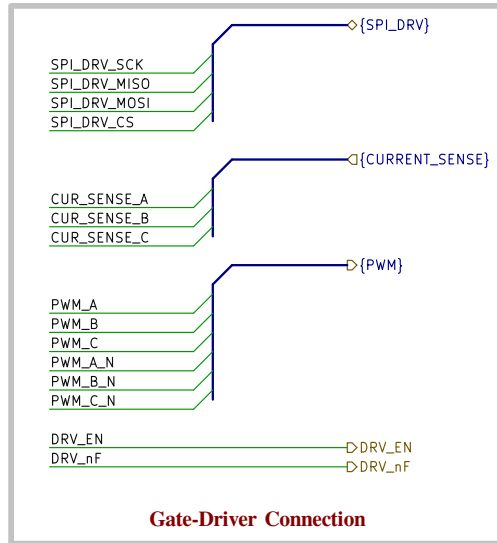
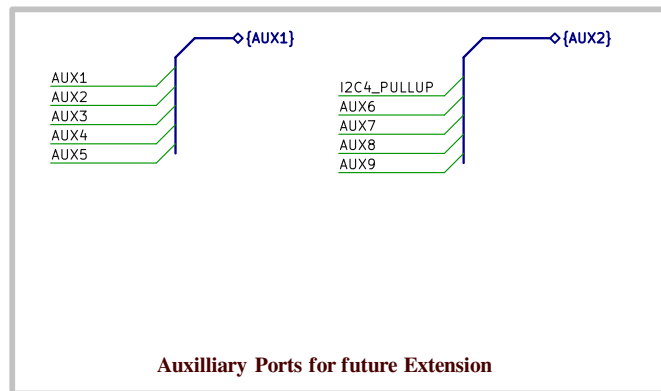
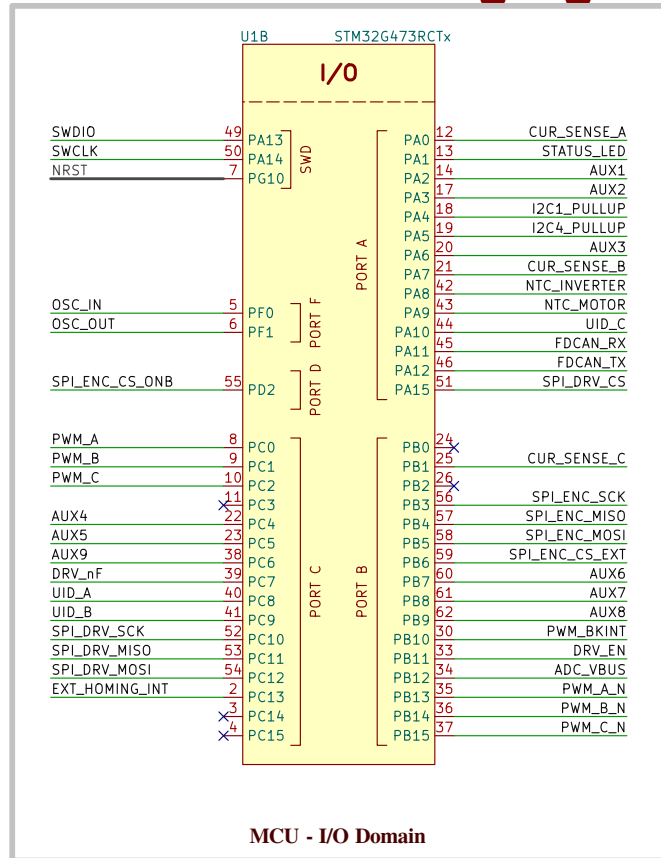
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	Board Name: ATLAS Motion			
	Sheet Title: Power	File Name: Power.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
	Sheet Path: /Power/		Size: A4	Sheet: 3 of 13

[4] MCU – Power



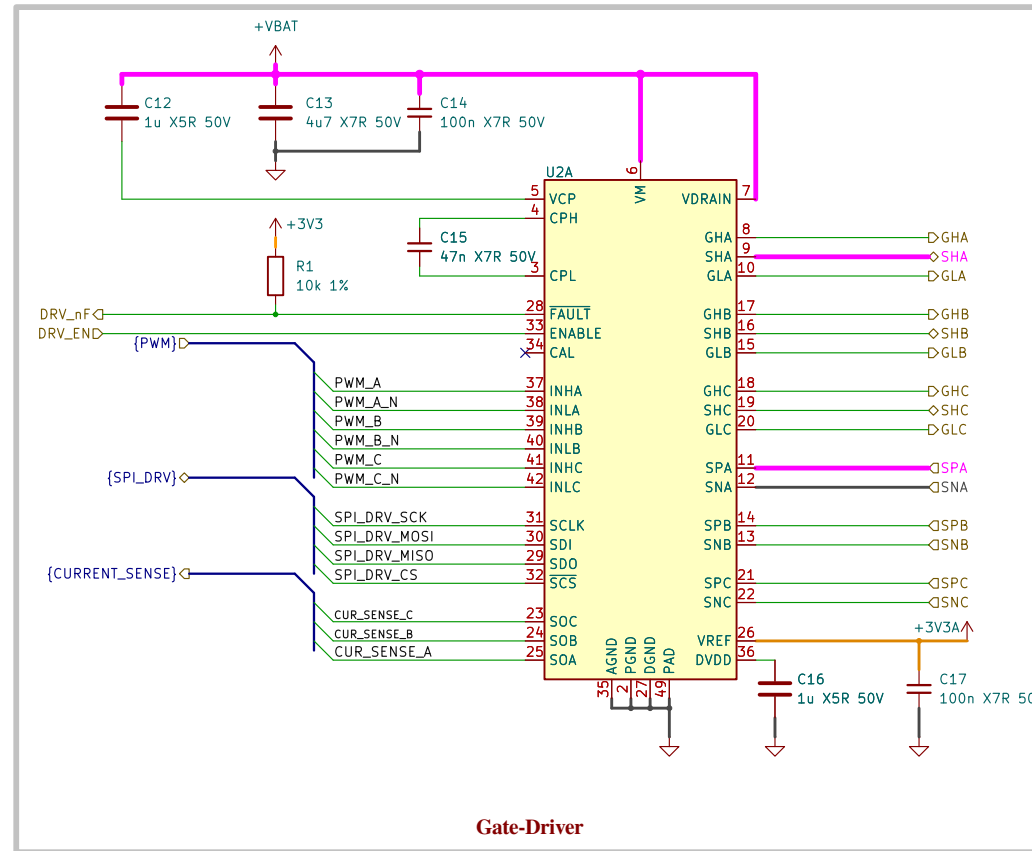
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	Board Name: ATLAS Motion			
	Sheet Title: MCU - Power	File Name: mcu-power.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
	Sheet Path: /MCU - Power/		Size: A4	Sheet: 4 of 13

[5] MCU – I/O Domain



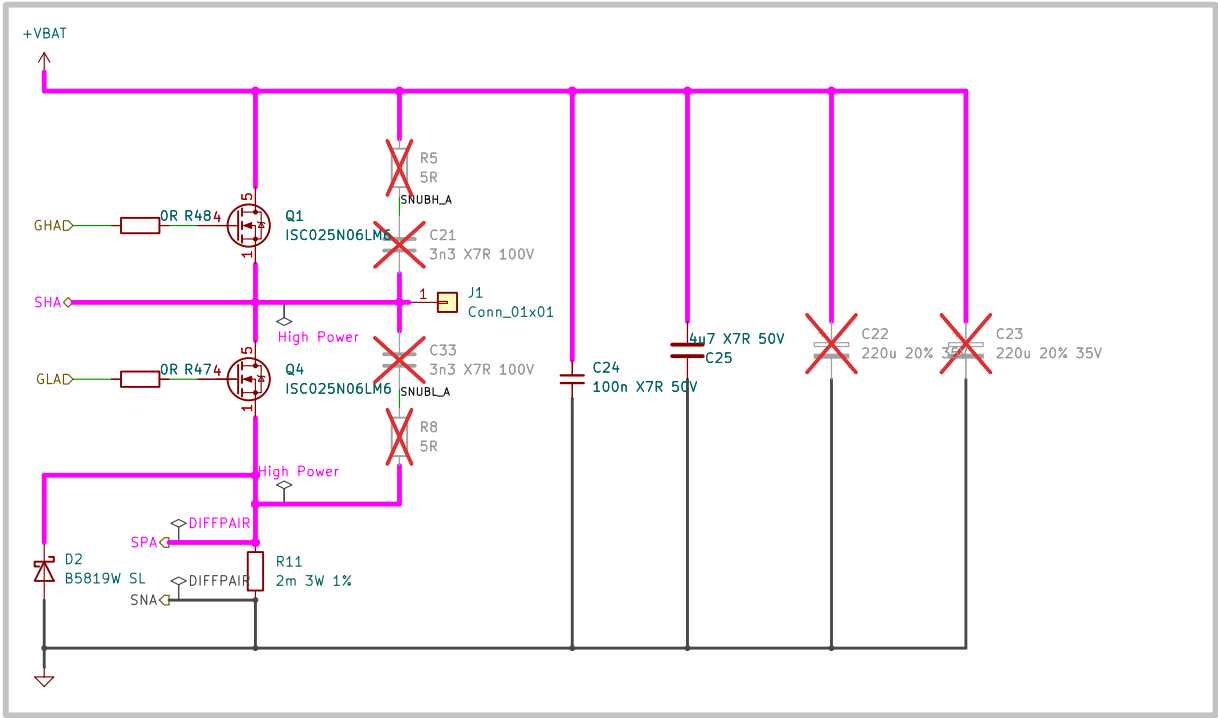
Comments:	Designer: Moritz Heinze	Date: 2026-09-07	Company: ATLAS ROBOTIC SYSTEMS	
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[6] FOC – Gate Driver



Comments:	Designer: Moritz Heinze	Date: 2026-09-07	ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
	Sheet Title: FOC - Gate Driver	File Name: gate-driver.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
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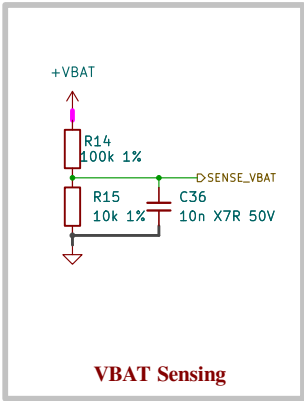
[7] FOC – Inverter



- GHAD — TP15 TestPoint
- SHA — TP16 TestPoint
- GLAD — TP17 TestPoint
- GHBD — TP18 TestPoint
- SHB — TP19 TestPoint
- GLBD — TP20 TestPoint
- GHCD — TP21 TestPoint
- SHC — TP22 TestPoint
- GHCD — TP23 TestPoint

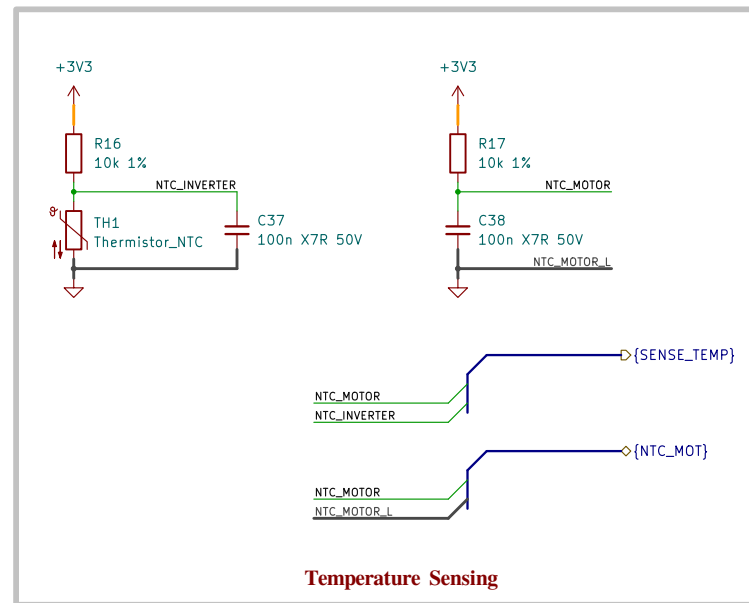
Comments:	Designer: Moritz Heinzle	Date: 2026-09-07	Company: ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion		Sheet Title:	File Name: inverter.kicad_sch
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[8] Sensing – VBAT



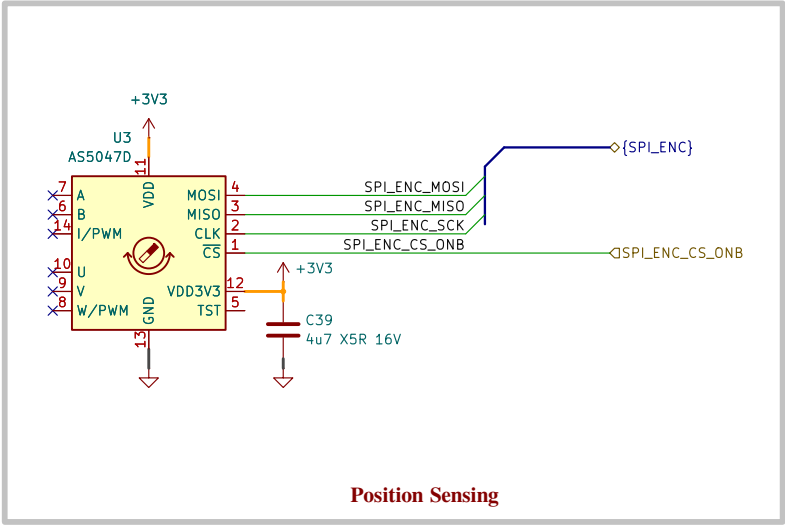
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	Board Name: ATLAS Motion			
	Sheet Title:	File Name: sense_temp.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
	Sheet Path: /Sensing - VBAT/		Size: A4	Sheet: 8 of 13

[9] Sensing – Temperature



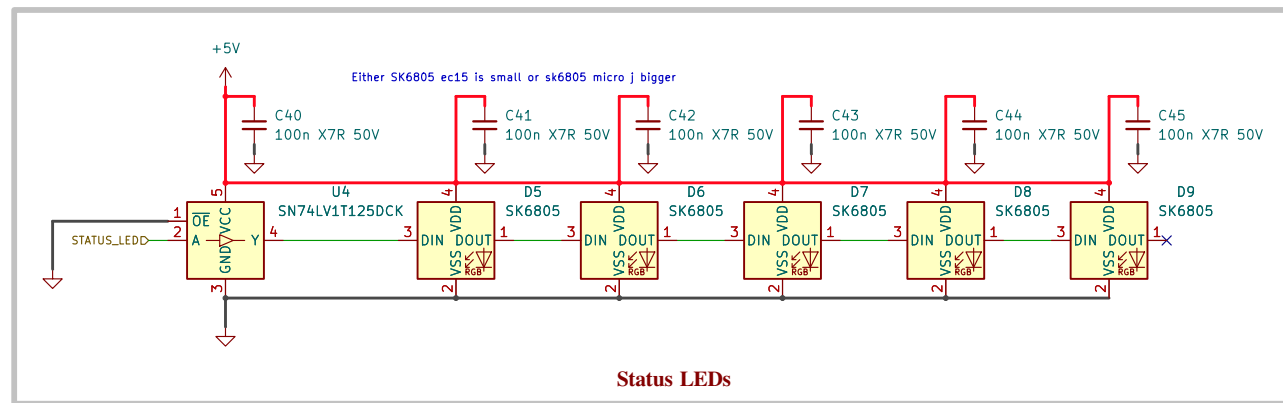
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	Board Name: ATLAS Motion			
	Sheet Title:	File Name: sense_vbat.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
	Sheet Path: /Sensing - Temperature/		Size: A4	Sheet: 9 of 13

[10] Sensing – Position



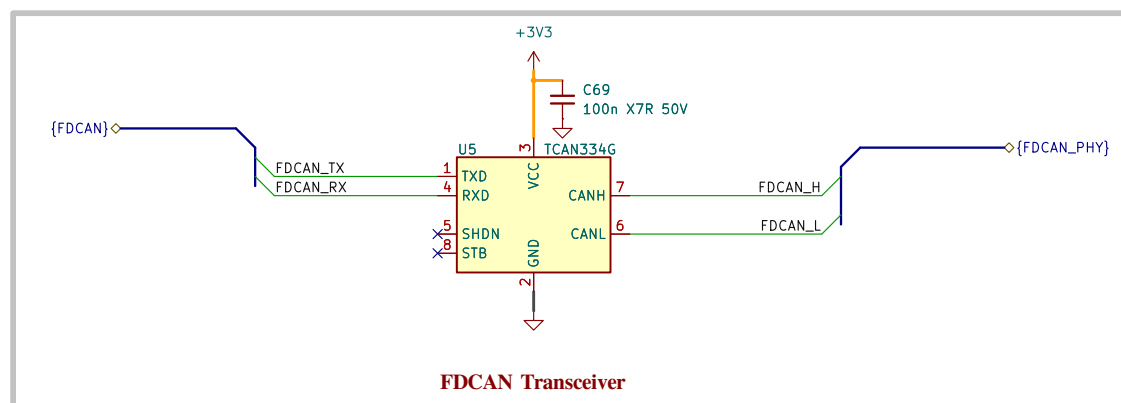
Comments:	Designer: Moritz Heinze	Date: 2026-09-07	ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
	Sheet Title:	File Name: sense_pos.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
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[11] Interface – Status LEDs



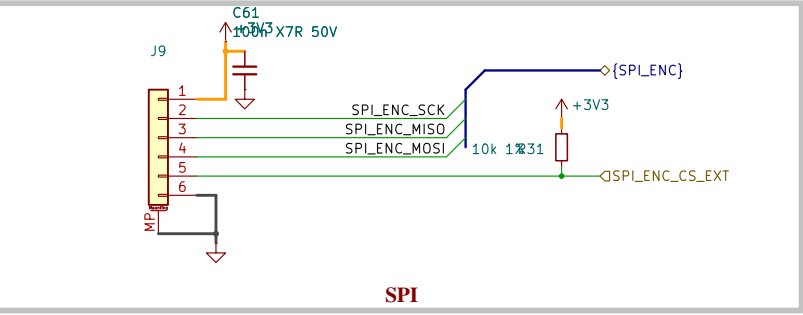
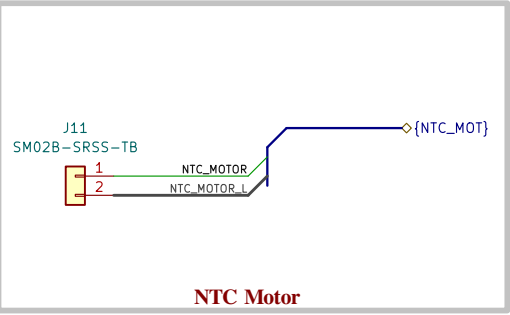
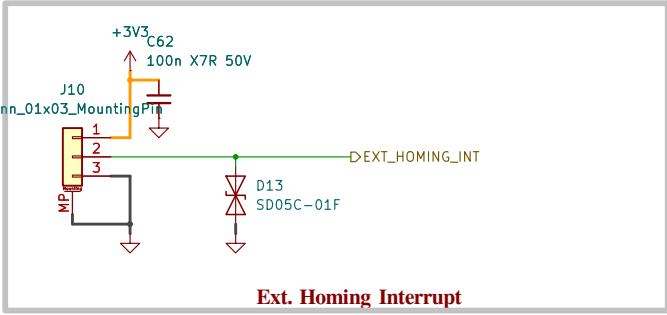
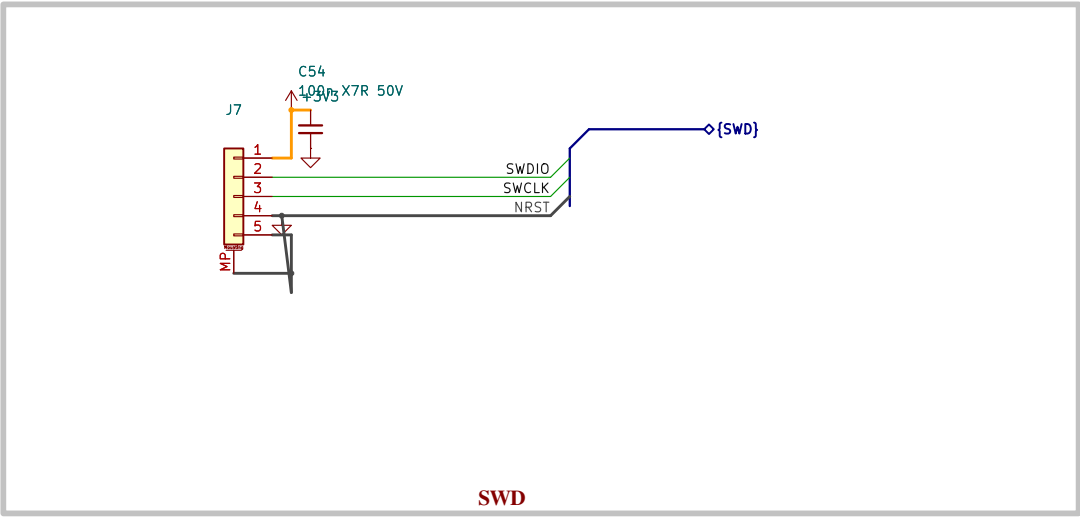
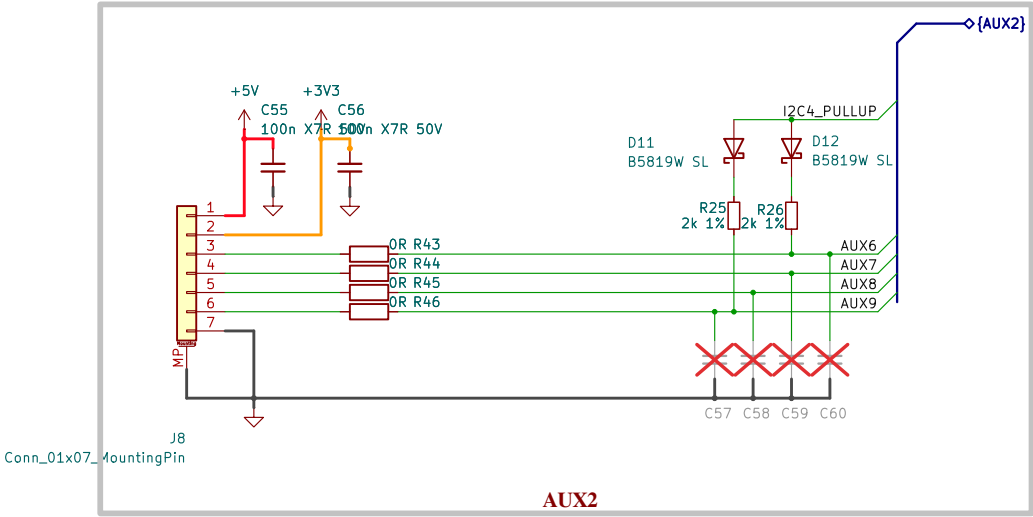
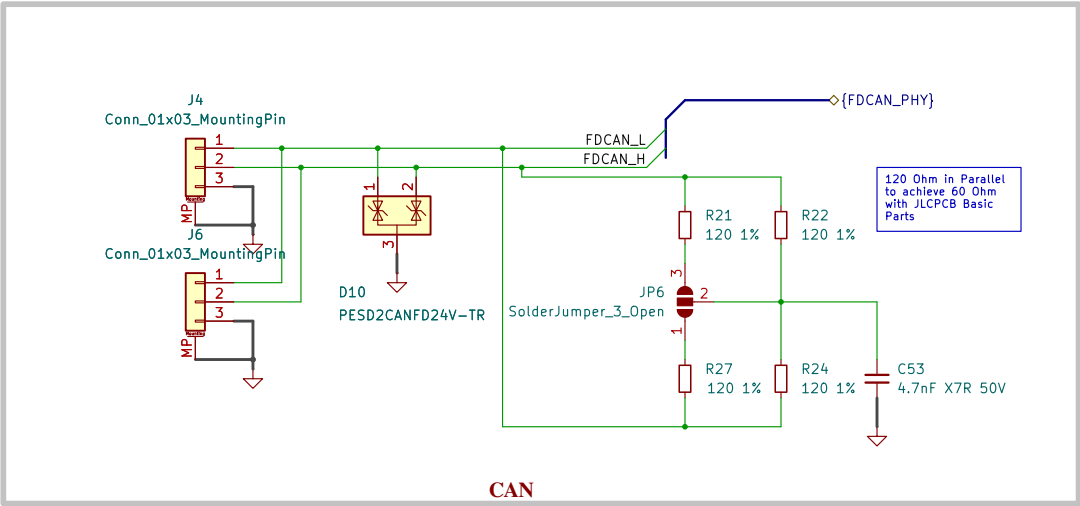
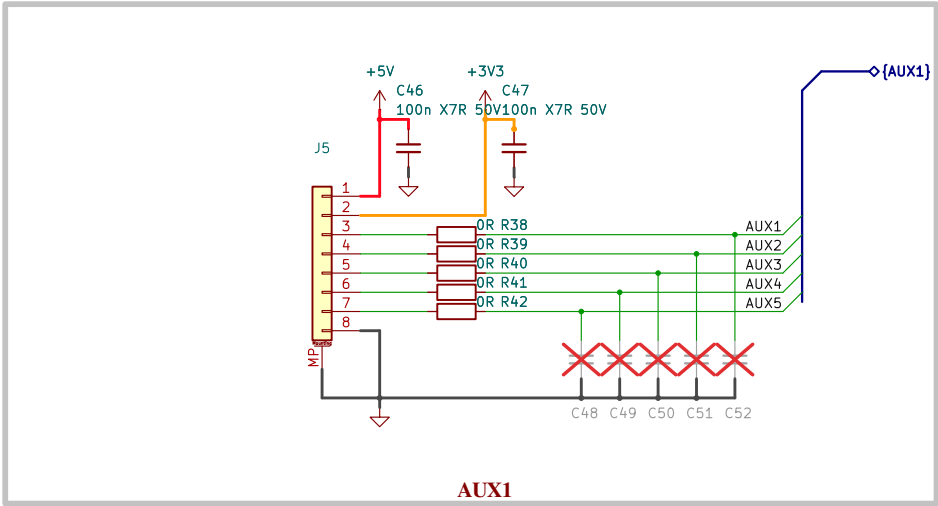
Comments:	Designer: Moritz Heinze	Date: 2026-09-07	ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
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	Sheet Path: /Interface - Status LEDs/		Size: A4	Sheet: 11 of 13

[12] Interface – FDCAN Tran.



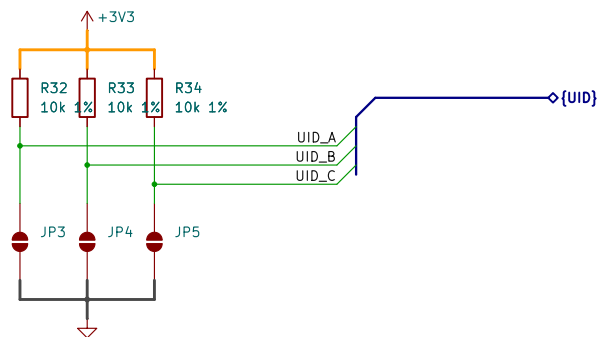
Comments:	Designer: Moritz Heinze	Date: 2026-09-07	ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
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	Sheet Path: /Interface - FDCAN Tran./		Size: A4	Sheet: 12 of 13

[13] Interface – Interconnects.

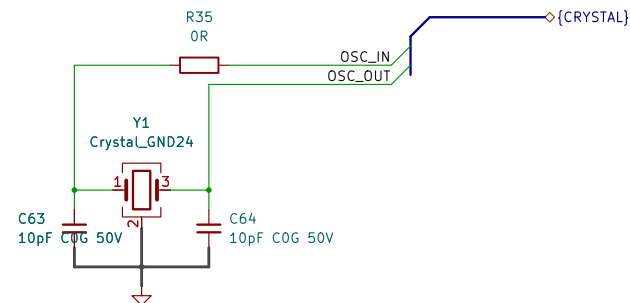


Comments:	Designer: Moritz Heinze	Date: 2026-09-07	ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
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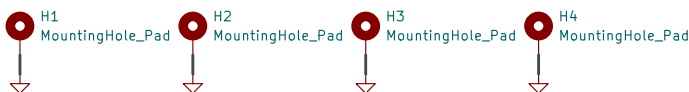
[13] MISC.



UID



Crystal



Mounting Holes

Comments:	Designer: Moritz Heinzle	Date: 2026-09-07	Company: ATLAS ROBOTIC SYSTEMS	
	Board Name: ATLAS Motion			
	Sheet Title:	File Name: misc.kicad_sch	Git Hash: as89s1	Revision: 0.0.1
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